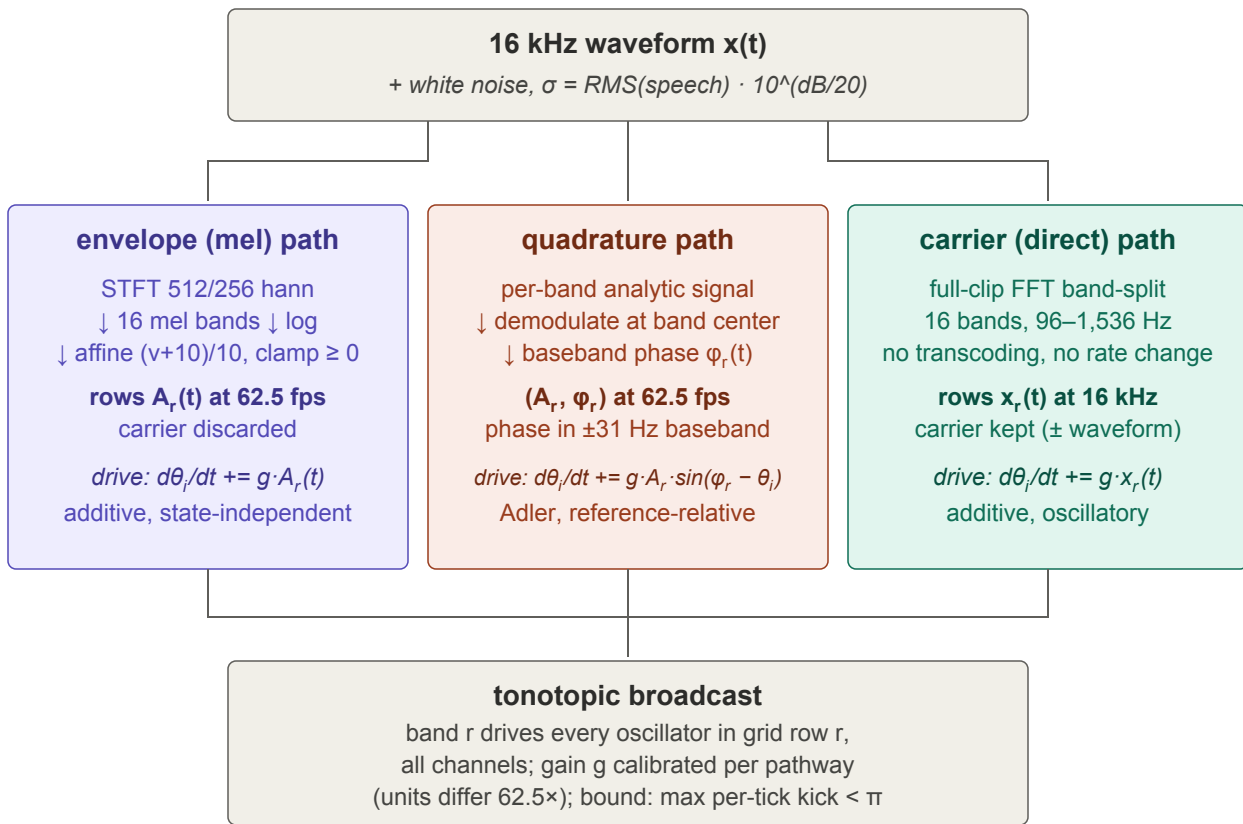




what each pathway excites in an untrained field (evidence-tagged)

envelope: state-independent pushes accumulate into readable loudness statistics
(measured: fields hold floor+3 across matrices A–C; weak envelope-rate locking observed, PLV ≈ 0.16)

quadrature: collapsed at every gain tested (measured, 64× sweep + matrices) — consistent with
Adler theory: reference-relative torque requires a consistent phase relation, which random
dynamics lack; trainability of that relation is suggested, not shown

carrier: oscillatory push admits injection locking when $|\Delta\omega| \leq A$ (Adler, single-oscillator
idealization; measured: drive-phase PLV rises with gain on frequency-matched rows)

information trade: envelope keeps 0–8 kHz energy at 16 ms resolution; carrier keeps phase, 96–1,536 Hz only